

GREYTOWN, South Africa

WMO: 684870

Lat: 29.083S

Lon: 30.603E

Elev: 1029

StdP: 89.56

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.5	2.8	-5.8	2.6	16.7	-3.7	3.1	15.5	7.3	15.8	6.3	15.8	0.6	10	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	32.7	19.7	31.0	19.4	29.5	19.1	23.1	28.4	22.2	27.2	21.4	26.1	3.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	18.5	25.6	20.7	17.5	24.5	20.0	16.7	23.6	74.4	28.3	70.7	27.1	67.5	26.1	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.5	4.9	-1.2	36.0	1.4	1.8	-2.2	37.2	-3.0	38.3	-3.8	39.2	-4.8	40.5		
(5)			-1.7	24.7	1.4	1.5	-2.6	25.7	-3.5	26.6	-4.2	27.4	-5.2	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	21.2	21.6	20.7	18.0	15.2	12.9	12.7	15.1	16.9	17.9	19.1	20.5
	DBStd	4.56	3.21	2.73	2.93	3.08	3.09	2.98	3.07	3.76	4.34	4.18	3.84	3.52
	HDD10.0	27	0	0	0	0	2	7	10	4	3	1	1	0
	HDD18.3	819	12	6	13	41	103	165	176	112	77	61	37	18
	CDD10.0	2813	346	323	331	242	164	93	93	163	210	246	274	327
	CDD18.3	564	99	96	86	33	7	2	1	12	34	48	61	87
	CDH23.3	3964	671	529	497	191	87	18	16	123	339	417	443	633
CDH26.7	1229	230	158	143	37	13	1	1	20	107	147	145	227	
Wind	WSAvg	1.9	2.0	1.8	1.7	1.5	1.4	1.5	1.6	2.0	2.2	2.3	2.4	2.1
Precipitation	PrecAvg	791	117	103	91	39	19	14	23	28	42	87	104	126
	PrecMax	1062	228	255	155	113	65	95	140	97	116	205	212	212
	PrecMin	486	36	37	48	7	0	0	0	1	5	17	33	53
	PrecStd	144	51	57	33	25	17	20	37	24	28	45	40	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	33.2	32.2	29.9	28.4	25.4	25.5	29.0	32.7	33.7	33.1	34.6
		MCWB	21.4	21.2	20.7	18.3	16.2	13.1	12.0	13.6	16.3	17.7	18.5	22.1
	2%	DB	31.5	30.5	30.1	27.5	25.7	22.9	23.0	26.5	29.7	30.6	30.5	31.4
		MCWB	20.9	21.1	20.1	17.7	15.2	12.1	11.2	13.2	15.2	16.9	18.9	20.6
	5%	DB	29.2	28.6	28.1	25.4	23.5	20.9	21.1	24.4	27.1	28.0	28.2	29.1
		MCWB	20.7	20.8	19.9	17.3	14.4	11.4	11.0	12.6	15.0	16.9	18.6	20.4
	10%	DB	27.1	26.6	26.2	23.5	21.4	18.9	19.0	22.1	24.7	25.4	25.8	26.9
		MCWB	20.5	20.6	19.6	16.9	13.7	10.8	10.6	12.2	14.5	16.6	18.4	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.1	23.2	21.4	18.2	15.9	14.2	16.9	19.1	20.6	22.2	23.8
		MCDB	29.8	29.6	28.3	25.9	24.2	20.1	19.6	22.8	27.4	28.6	28.0	31.0
	2%	WB	22.7	22.9	21.9	19.6	16.8	13.9	13.2	15.4	17.4	19.2	20.8	22.5
		MCDB	27.9	27.5	27.0	23.9	22.4	19.8	18.7	21.7	25.1	26.0	26.6	28.4
	5%	WB	21.6	22.0	20.9	18.6	15.7	12.8	12.4	14.4	16.3	18.2	20.0	21.3
		MCDB	26.5	26.0	25.5	22.8	21.1	18.1	17.8	20.1	23.1	24.6	25.4	26.7
	10%	WB	20.7	21.1	20.1	17.7	14.7	11.9	11.5	13.5	15.4	17.4	19.1	20.4
		MCDB	25.5	25.0	24.5	21.6	19.5	17.0	16.9	18.9	21.6	23.1	24.1	25.3
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.3	10.0	10.8	13.3	13.4	13.7	12.6	12.4	10.6	10.0	9.9
		MCDBR	14.2	13.0	13.6	14.1	16.2	15.6	15.9	15.9	16.8	16.5	14.8	14.6
		MCWBR	5.5	4.9	5.3	5.3	6.8	6.7	6.9	5.6	5.5	5.5	5.4	5.7
	5% WB	MCDBR	12.1	11.2	11.9	11.3	14.2	12.7	13.4	12.5	14.1	13.7	12.6	12.6
		MCWBR	5.8	5.4	5.7	5.1	6.6	6.5	6.9	5.3	5.8	5.6	5.9	6.1
Clear-Sky Solar Irradiance	taub	0.394	0.385	0.377	0.347	0.319	0.302	0.310	0.367	0.412	0.432	0.403	0.400	
	taud	2.368	2.395	2.400	2.443	2.485	2.508	2.474	2.317	2.216	2.203	2.314	2.340	
	Ebn at Noon	948	938	913	892	873	868	874	855	863	883	933	945	
	Edh at Noon	131	124	117	101	87	80	87	113	138	149	138	136	
All-Sky Solar Radiation	RadAvg	5.97	5.87	5.18	4.29	3.75	3.36	3.60	4.20	4.82	5.09	5.63	5.98	
	RadStd	0.43	0.47	0.29	0.27	0.20	0.17	0.21	0.26	0.30	0.45	0.48	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page